

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0872

Revere 100 kV
August 23, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0872. The Interconnection Customer proposes a 43.7 MW natural gas combined cycle facility interconnecting at the Revere 100 kV substation in the Indiana area. The request was received on October 07, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 401 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$412,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Twin Branch - Freehaven-Lakepark 100.0 kV L	Whitestone - Stepstone 100.0 kV line	119.3	219	34.3
Segs 7 - Freehaven-Lakepark 100.0 kV Cir. 1	Freehaven-Lakepark - Segs 7 100.0 kV #2	117.7	214	33.8
Kemper County - Twin Branch 100.0 kV	Stepstoe - Whitestone 100.0 kV	116.2	246	34.3
German - Colfax 100.0 kV	Drewry - Colfax 100.0 kV	106.2	111	12.4
Concord - Segs 7 100.0 kV Line 1	Segs 7 - Concord 100.0 kV Cir. 2	102.3	224	33.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Freehaven-Lakepark - Twin Branch 100.0 kV	Rebuild line with higher-capacity conductor	\$2,759,000	4% (\$106,000)
Segs 7 - Freehaven-Lakepark 100.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,431,000	3% (\$80,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Twin Branch - Kemper County Substation 100.0 kV	Rebuild line with higher-capacity conductor	\$4,350,000	4% (\$167,000)
Colfax - German Substation 100.0 kV	Reconductor line	\$1,006,000	4% (\$40,000)
Concord - Seg 7 100.0 kV #1	Reconductor line	\$576,000	3% (\$19,000)
	Total allocated to project		\$412,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.